Computer Science Portfolio Report

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Contents

Abstract

This report describes the design, structure, implementation, and future development of my computer science portfolio. The portfolio was created to showcase my technical skills, academic progress, and professional interests as I prepare for working student opportunities in software engineering. It consists of a responsive portfolio website, a GitHub repository containing the source code and supporting documentation, and a curriculum vitae created using LaTeX. This report explains the organisation of these components, the technologies used, the design decisions made during development, and areas for future improvement.

1 Introduction

A professional portfolio is an important tool for software engineering students because it provides a structured way to present technical skills, academic achievements, and career goals. While a curriculum vitae summarizes qualifications and experience, a portfolio offers a more detailed view of a student's work, technical abilities, and approach to organisation and documentation.

The purpose of this project was to develop a computer science portfolio that can be shared with potential employers when applying for working student positions. The portfolio demonstrates not only technical skills but also the ability to organise projects, maintain documentation, and publish content professionally.

The portfolio website is available online through GitHub Pages:

https://aiwyn-007.github.io/Portfolio/

The source code and supporting materials are available in the GitHub repository:

https://github.com/Aiwyn-007/Portfolio

The portfolio presents my profile as a Software Engineering student based in Berlin. It reflects my interests in software development, web technologies, databases, algorithms, and user-centred solutions. The project also demonstrates practical use of GitHub for version control and deployment, as well as LaTeX for producing professional documentation.

2 Portfolio Website Structure

2.1 Website Purpose and Content

The website was designed as a single-page portfolio. This approach was chosen because it allows visitors to access the most relevant information quickly without navigating through multiple pages. The navigation menu uses anchor links that enable users to move directly between sections.

The website contains the following main sections:

- **Introduction Section** – Provides a brief overview of my background, interests, and professional goals.

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- **About Section** – Describes my approach to learning, teamwork, and problem-solving within software engineering.
- **Skills Section** – Highlights programming languages, web development fundamentals, technical tools, and professional competencies.
- **Education Section** – Presents information about my Software Engineering degree and expected graduation timeline.
- **Portfolio Section** – Introduces the purpose of the portfolio and outlines plans for future project additions.
- **Contact Section** – Provides contact information along with links to my GitHub repository and downloadable CV. “‘

The content is intentionally concise and easy to scan. Since recruiters and hiring managers often spend only a short time reviewing applications, the website uses short paragraphs, headings, and structured lists to improve readability.

2.2 Visual Design Decisions

The website uses a dark red and black colour scheme to create a modern and professional appearance. The dark background contributes to a technical aesthetic, while red is used as an accent colour for headings, buttons, and interactive elements. This helps important information stand out while maintaining a consistent visual identity.

The layout incorporates whitespace, rounded content cards, and clearly separated sections. These design choices improve readability and allow visitors to navigate the content more easily.

Responsive design techniques were implemented using CSS Grid layouts and media queries. As a result, the website adapts effectively to different screen sizes and remains accessible on both desktop and mobile devices.

At its current stage, the website intentionally focuses on textual information rather than visual content. Since the primary objective is to present personal information, skills, education, and documentation, extensive imagery is not yet required. Future versions may include screenshots, diagrams, and project demonstrations as additional work is completed.

3 GitHub Repository Structure

The repository is organised to separate website files from supporting documents and source materials. The main structure is shown below:

```
.
 index.html
 README.md
 LICENSE
 assets/
    images/
    documents/
```

```
        Aiwyn_Anish_CV.pdf
cv/
   cv.tex
report/
portfolio-report.tex
portfolio-report.pdf
```

The `index.html` file serves as the main entry point of the website and is used by GitHub Pages to display the live portfolio.

The `README.md` file provides an overview of the repository, including its purpose, structure, and technologies.

The `assets` directory stores resources used by the website. The `documents` folder contains the PDF version of my CV, while the `images` folder is reserved for future screenshots, diagrams, and profile images.

The `cv` directory contains the LaTeX source code used to generate my curriculum vitae. The `report` directory contains the source files and compiled PDF version of this report.

This structure was chosen because it is easy to understand, maintain, and navigate. It keeps related files grouped together while ensuring that visitors can quickly locate important resources.

4 Tools and Technologies

Several technologies were used throughout the development of the portfolio:

- HTML for structuring website content.
- CSS for styling, responsive layouts, typography, and visual design.
- Git and GitHub for version control, repository hosting, and project management.
- GitHub Pages for publishing the portfolio website.
- LaTeX for creating the CV and project report.

GitHub Pages was selected because it provides a simple and reliable solution for hosting static websites. It also allows the live website to remain directly connected to the source repository, making the implementation transparent to potential employers.

LaTeX was chosen for creating both the CV and this report because it provides professional formatting, consistency, and reproducible document generation. Including the source files in the repository also demonstrates good documentation practices.

5 Strengths and Limitations

One of the portfolio's main strengths is its clear organisation. The website, documentation, CV, and supporting files are all stored within a single repository using a logical directory structure. This makes it easier for employers or reviewers to locate relevant information.

Another strength is the responsive design of the website. Flexible layouts and media queries ensure that content remains accessible across a range of devices. The portfolio also maintains a consistent visual identity and a clear hierarchy of information.

The project demonstrates both technical and professional skills. In addition to web development, it highlights experience with version control, documentation, and the use of LaTeX for producing structured technical documents.

However, the portfolio is still at an early stage of development. At present, it focuses primarily on personal information, education, and technical skills rather than completed software projects. While this reflects my current stage of study, the portfolio will become stronger as additional projects are completed.

A further limitation is the relatively simple level of interactivity. The website was intentionally designed to remain lightweight and easy to maintain. Future enhancements could introduce more dynamic content and interactive demonstrations.

6 Future Improvements

There are several opportunities to improve the portfolio as my skills and experience continue to develop.

One priority is the addition of completed coursework and software projects. Future project entries should include problem descriptions, technical decisions, implementation details, setup instructions, and reflections on lessons learned.

Another improvement would be the inclusion of screenshots, diagrams, and visual demonstrations. These elements would help communicate technical concepts more effectively and make the website more engaging.

Additional enhancements may include links to professional networking platforms, improved CV accessibility, and more detailed descriptions of technical skills and experience.

The website could also be expanded with dedicated project pages and a secure contact form, providing a more comprehensive professional presence.

7 Conclusion

This project resulted in the creation of a structured and professional computer science portfolio designed for working student applications. The portfolio combines a responsive GitHub Pages website, a well-organised GitHub repository, a LaTeX-generated CV, and supporting documentation.

The portfolio provides a strong foundation for presenting my skills, academic progress, and professional interests as a Software Engineering student. As I complete new coursework and develop additional software projects, the portfolio will continue to evolve and become a more comprehensive representation of my abilities and experience.